Fstab Stabilization - UUID Conversion

Overview

After migrating VMs from VMware/Hyper-V to KVM, some VMs may have fstab entries that reference devices using unstable identifiers like `/dev/disk/by-path/pci-*`. These identifiers are hardware-specific and will break when the VM is migrated to different hardware or when PCI addresses change in the virtualized environment.

This document describes the process of converting all fstab entries to use stable UUID identifiers.

Problem Statement

Unstable Device Identifiers

Some distributions (particularly openSUSE) use PCI-based device paths:

```
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2
```

These paths are problematic because: - They are tied to specific PCI slot addresses - They change when VM hardware configuration changes - They break when migrating between hypervisors - They fail in KVM when the virtual hardware layout differs from the source

Stable Alternative: UUID

UUID (Universally Unique Identifier) provides a stable way to reference filesystems:

```
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85
```

Benefits: - Unique to the filesystem, not the hardware - Remains constant across migrations - Works regardless of PCI addressing - Standard practice for modern Linux systems

Solution Approach

Process

1. **Shutdown VMs**: Release disk locks
2. **Connect Disk via NBD**: Use qemu-nbd to access disk images offline
3. **Detect Partition Layout**: Handle GPT, MBR, LVM, and Btrfs subvolumes
4. **Extract UUIDs**: Use `blkid` to get filesystem UUIDs
5. **Update fstab**: Replace unstable identifiers with UUIDs
6. **Verify Boot**: Start VMs and confirm successful boot

Tools Used

- `qemu-nbd` : Network Block Device for offline disk access
- `blkid` : UUID extraction
- `lvm2` : LVM volume group management
- `btrfs-progs` : Btrfs subvolume mounting

VM-Specific Results

1. openSUSE Leap 15.4

Status: FIXED (11 entries converted)

Original fstab (by-path):

```
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /
btrfs  defaults                      0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /var
btrfs  subvol=/@/var                 0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /usr/local
btrfs  subvol=/@/usr/local           0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /tmp
btrfs  subvol=/@/tmp                 0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /srv
btrfs  subvol=/@/srv                 0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /root
btrfs  subvol=/@/root                0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /opt
btrfs  subvol=/@/opt                 0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /home
```

```
btrfs  subvol=/@/home                0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /boot/grub2/x86_64-efi
btrfs  subvol=/@/boot/grub2/x86_64-efi  0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /boot/grub2/i386-pc
btrfs  subvol=/@/boot/grub2/i386-pc  0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part2  /.snapshots
btrfs  subvol=/@/.snapshots          0  0
/dev/disk/by-path/pci-0000:00:10.0-scsi-0:0:0:0-part3  swap
swap   defaults                      0  0
```

Updated fstab (UUID):

```
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /                       btrfs
defaults                      0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /var                    btrfs
subvol=/@/var                 0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /usr/local              btrfs
subvol=/@/usr/local           0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /tmp                    btrfs
subvol=/@/tmp                 0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /srv                    btrfs
subvol=/@/srv                 0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /root                   btrfs
subvol=/@/root                0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /opt                    btrfs
subvol=/@/opt                 0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /home                   btrfs
subvol=/@/home                0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /boot/grub2/x86_64-efi  btrfs
subvol=/@/boot/grub2/x86_64-efi  0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /boot/grub2/i386-pc     btrfs
subvol=/@/boot/grub2/i386-pc  0  0
UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85  /.snapshots             btrfs
subvol=/@/.snapshots          0  0
UUID=03c038b5-fb29-470c-9f81-7100da936770  swap                    swap
defaults                      0  0
```

Partition Layout: - `/dev/nbd14p1` : 8M BIOS boot partition - `/dev/nbd14p2` : 510G Btrfs root (UUID: f293ef3c-255a-4582-8016-f72fb8dd3f85) - `/dev/nbd14p3` : 2G swap (UUID: 03c038b5-fb29-470c-9f81-7100da936770)

Boot Status: Running successfully

---

2. Ubuntu 25 Server

Status: FIXED (1 entry converted)

Original fstab (dm-uuid):

```
# / was on /dev/ubuntu-vg/ubuntu-lv during curtin installation
/dev/disk/by-id/dm-uuid-LVM-
TjdITyfq4bzjFZ72abuWRyAEBfhingPTcj6PMKRuLQ93CevoO4kJDDKVit8KAQbX / ext4 defaults
0 1
# /boot was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/6f0d41a2-60a5-4400-bfb6-c384032694f9 /boot ext4 defaults 0 1
/swap.img   none    swap    sw  0   0
```

Updated fstab (UUID):

```
# / was on /dev/ubuntu-vg/ubuntu-lv during curtin installation
UUID=90bc13b5-9f1d-4865-92a9-cc87d93385ca / ext4 defaults 0 1
# /boot was on /dev/sda2 during curtin installation
UUID=6f0d41a2-60a5-4400-bfb6-c384032694f9 /boot ext4 defaults 0 1
/swap.img   none    swap    sw  0   0
```

Partition Layout: - `/dev/nbd14p1` : 1M BIOS boot partition - `/dev/nbd14p2` : 2G ext4 boot (UUID: 6f0d41a2-60a5-4400-bfb6-c384032694f9) - `/dev/nbd14p3` : 498G LVM physical volume - VG: `ubuntu-vg` - LV: `ubuntu-lv` (100G ext4, UUID: 90bc13b5-9f1d-4865-92a9-cc87d93385ca)

Boot Status: Running successfully

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3. Fedora Cloud 43

Status: ALREADY STABLE (no changes needed)

Existing fstab (already using UUID):

```
UUID=55773572-92c8-4d7e-bafd-3f4943a2f380 / btrfs
compress=zstd:1,defaults,subvol=root 0 1
UUID=fc8240da-f3f1-47a3-87a1-4e80eb1371d0 /boot ext4 defaults 0 0
UUID=55773572-92c8-4d7e-bafd-3f4943a2f380 /home btrfs
compress=zstd:1,subvol=home 0 0
UUID=55773572-92c8-4d7e-bafd-3f4943a2f380 /var btrfs compress=zstd:1,subvol=var
```

```
0 0
UUID=980C-1628 /boot/efi vfat defaults,umask=0077,shortname=winnt 0 0
```

Partition Layout: - `/dev/nbd14p1` : 2M BIOS boot partition - `/dev/nbd14p2` : 100M ext4 boot (UUID: fc8240da-f3f1-47a3-87a1-4e80eb1371d0) - `/dev/nbd14p3` : 2G vfat EFI (UUID: 980C-1628) - `/dev/nbd14p4` : 2.9G Btrfs root (UUID: 55773572-92c8-4d7e-bafd-3f4943a2f380)

Boot Status: Running successfully

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4. CentOS 10 Server

Status: ALREADY STABLE (no changes needed)

Existing fstab (already using UUID):

```
UUID=3e0ad6b4-20e0-424d-999f-ce2e70cfb86e /                       xfs
defaults        0 0
UUID=15547891-2d55-4522-b27a-2183461fd641 /boot                   xfs
defaults        0 0
UUID=61907098-d80a-48ac-8b66-fffaa3d45bc5 /home                   xfs
defaults        0 0
UUID=54811d97-6965-4156-9527-f6cb43b70f2f none                    swap
defaults        0 0
```

Partition Layout: - `/dev/nbd4p1` : 1M BIOS boot partition - `/dev/nbd4p2` : 1G XFS boot (UUID: 15547891-2d55-4522-b27a-2183461fd641) - `/dev/nbd4p3` : 499G LVM physical volume - VG: `cs` - LVs: - `root` : 70G XFS (UUID: 3e0ad6b4-20e0-424d-999f-ce2e70cfb86e) - `home` : 424.18G XFS (UUID: 61907098-d80a-48ac-8b66-fffaa3d45bc5) - `swap` : 4.81G (UUID: 54811d97-6965-4156-9527-f6cb43b70f2f)

Boot Status: Running successfully

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5. Ubuntu VMware

Status: ALREADY STABLE (no changes needed)

Existing fstab (already using UUID):

```
# / was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/d7da8f9b-5333-48d1-a96f-5c6ef03997ac / ext4 defaults 0 1
/swap.img    none     swap     sw  0    0
```

Partition Layout: - `/dev/nbd14p1` : 1M BIOS boot partition - `/dev/nbd14p2` : 20G ext4 root (UUID: d7da8f9b-5333-48d1-a96f-5c6ef03997ac)

Boot Status: Running successfully

Technical Notes

Handling Btrfs Subvolumes

For Btrfs filesystems with subvolumes (like openSUSE and Fedora): 1. Mount with specific subvolume: `mount -o subvol=root /dev/device /mnt` 2. All subvolumes share the same UUID 3. Subvolume names are specified in mount options: `subvol=/@/var`

Handling LVM

For LVM-based systems (like CentOS and Ubuntu 25): 1. Scan for volume groups: `vgscan` 2. Activate volume groups: `vgchange -ay` 3. Mount logical volumes: `mount /dev/vgname/lvname /mnt` 4. Deactivate before disconnecting: `vgchange -an vgname`

Important: The NBD device number must match the one LVM expects. If LVM shows device mismatch warnings, reconnect to the correct NBD device.

Device Mismatch Warning

When working with LVM, you may see:

```
WARNING: Device mismatch detected for cs/swap which is accessing /dev/nbd4p3
instead of /dev/nbd14p3.
```

Solution: Disconnect and reconnect to the NBD device LVM expects:

```
sudo vgchange -an vgname
sudo qemu-nbd -d /dev/nbd14
```

```
sudo qemu-nbd -c /dev/nbd4 /path/to/image.qcow2
sudo vgchange -ay
```

UUID vs by-uuid Notation

Both notations are valid in fstab: - `UUID=f293ef3c-255a-4582-8016-f72fb8dd3f85` - `/dev/disk/by-uuid/f293ef3c-255a-4582-8016-f72fb8dd3f85`

The `UUID=` notation is preferred as it's more concise.

Verification

All 5 VMs are now running with stable UUID-based fstab entries:

```
$ sudo virsh list --all
 Id   Name                   State
-----------------------------------
 1    opensuse-leap-15.4     running
 2    fedora43-cloud         running
 3    centos10-server        running
 4    ubuntu25-server        running
 5    ubuntu-vmware          running
```

Recommendations

For Future Migrations

1. **Always use fstab_mode: stabilize-all** in migration configs:

   ```
   fstab_fixes_enable: true
   fstab_mode: stabilize-all
   ```

2. **Verify fstab after migration**:

   ```
   sudo qemu-nbd -c /dev/nbd0 image.qcow2
   sudo partprobe /dev/nbd0
   sudo mount /dev/nbd0pX /mnt
   sudo cat /mnt/etc/fstab
   ```

3. **Test boot before production**: Always test boot in a development environment before deploying to production.

Best Practices

- Use UUID for all filesystem references
- Avoid device names like `/dev/sda1` (order can change)
- Avoid PCI paths (hardware-specific)
- For LVM, UUID is better than `/dev/vgname/lvname`
- For Btrfs, use UUID with subvolume options

Summary

VM	Original Identifier	Fixed	Boot Status
openSUSE Leap 15.4	by-path (PCI)		Running
Ubuntu 25 Server	dm-uuid (LVM)		Running
Fedora Cloud 43	UUID (stable)	N/A	Running
CentOS 10 Server	UUID (stable)	N/A	Running
Ubuntu VMware	UUID (stable)	N/A	Running

Result: 100% of VMs now use stable UUID identifiers and boot successfully in KVM.